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## Questions

**Q1.**

This question is about bonding.

(a) Draw an electron density map for a molecule of oxygen.

**(1)**

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(b) Draw a diagram to show the shape of a water molecule.

Give the bond angle.

**(2)**

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Bond angle .....

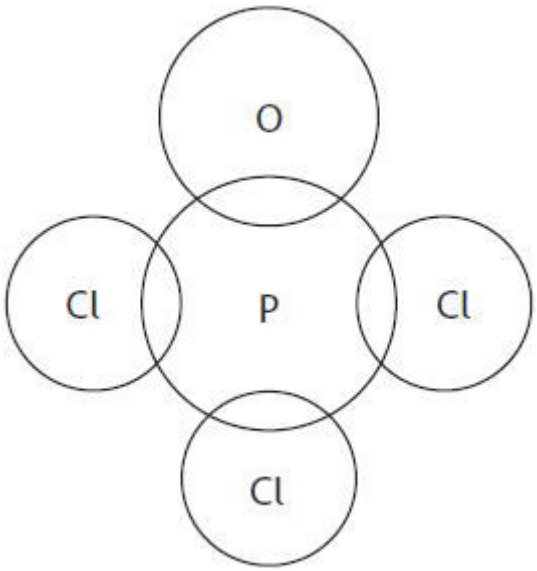
(c) The compound $POCl_3$ has a simple molecular structure.

(i) Complete the dot-and-cross diagram for the $POCl_3$ molecule.
Use crosses (✕) for the phosphorus electrons, dots (●) for the chlorine electrons and circles (○) for the oxygen electrons.

**(2)**

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(ii) Explain the shape of this molecule using the electron-pair repulsion theory.

**(3)**

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(d) The properties of metals depend on their structure and bonding.

(i) Draw a labelled diagram to show the metallic bonding in calcium.

**(2)**

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(ii) Explain how the electrical conductivity, high melting temperature and malleability of metals depend on their structure and bonding.

**(3)**

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Electrical conductivity.....

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High melting temperature .....

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Malleability .....

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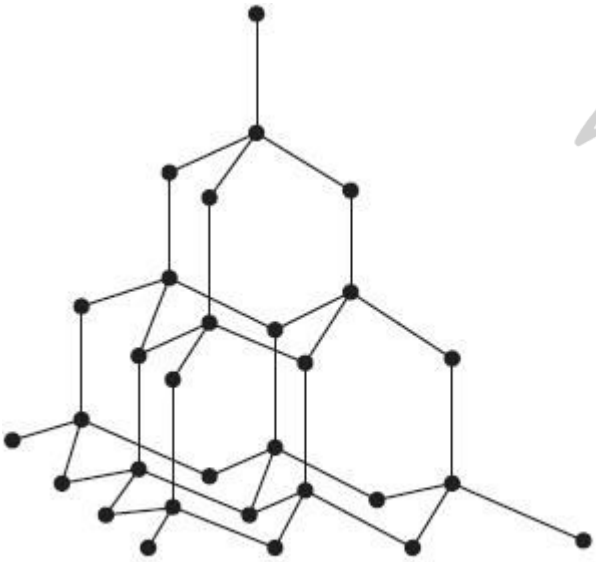
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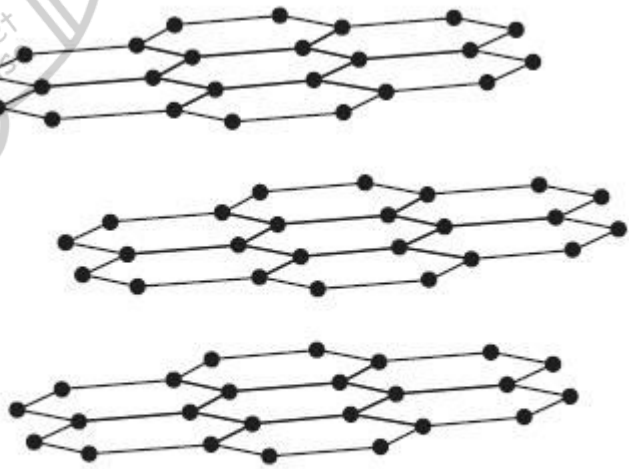
(e) Diamond, graphite and graphene are all forms of carbon.

(i) Explain **two** ways in which the physical properties of diamond and graphite differ. Refer to their structure and bonding in your answer.

**(4)**



diamond



graphite

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(ii) State how the structure of graphene is related to the structure of graphite.

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(1)

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(iii) State a use for graphene, identifying the property that makes it suitable for that use.

(2)

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**(Total for question = 20 marks)**

**Q2.**

Metallic bonding is the strong electrostatic attraction between

☒ **A** anions and cations

☒ **B** atoms and delocalised electrons

☒ **C** ions and delocalised electrons

☒ **D** two nuclei and a shared pair of electrons

**(Total for question = 1 mark)**

**Q3.**

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**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Covalent bonding is best described as the electrostatic attraction between

☒ **A** oppositely charged ions

☒ **B** positive ions and delocalised electrons

☒ **C** a shared pair of electrons

☒ **D** two nuclei and a shared pair of electrons

**(Total for question = 1 mark)**

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**Q4.**

This question is about the elements in Period 3 of the Periodic Table, and some of their compounds.

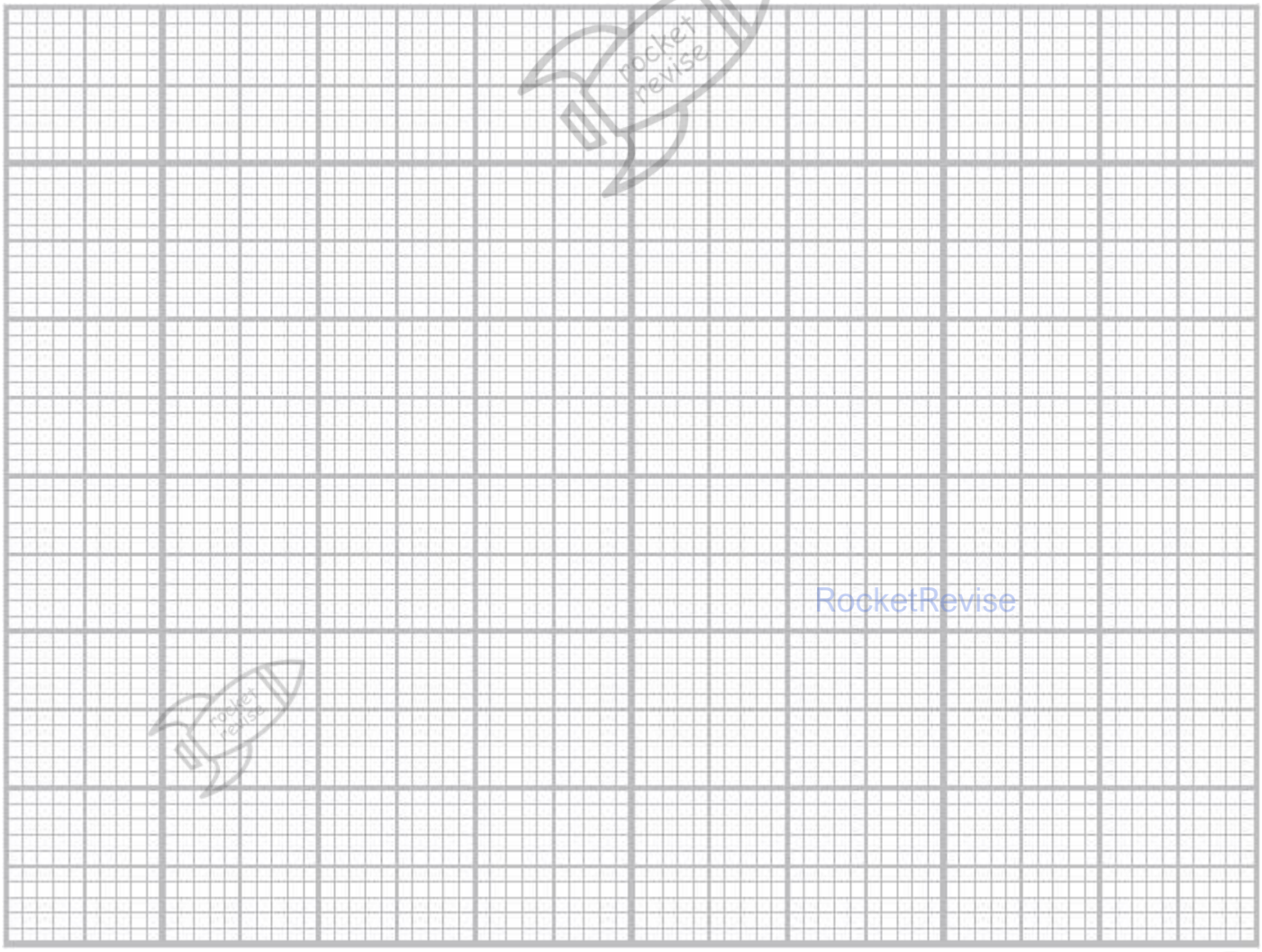
(a) The atomic radii of six of the elements are given.

| Symbol | Na | Mg | Al | Si | P | S | Cl | Ar |
|---|---|---|---|---|---|---|---|---|
| Atomic number | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| Atomic radius / nm | 0.191 | 0.160 | 0.130 | | | 0.102 | 0.099 | 0.095 |

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(i) Plot a graph of atomic radius against atomic number.

**(2)**




(ii) Use the graph to estimate the atomic radius of silicon, Si.

**(1)**

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(iii) Suggest an explanation for the decrease in atomic radius as atomic number increases across a period.

**(3)**

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(b) The melting temperatures of sodium, sodium chloride and chlorine are given in the table.

Complete the table to show the type of structure, the type of bond or force broken on melting and the particles involved.

**(6)**

| Substance | Sodium | Sodium chloride | Chlorine |
|---|---|---|---|
| Melting temperature / °C | 98 | 801 | −101 |
| Type of structure | giant | | simple molecular |
| Type of bond or force broken on melting | | | |
| Particles involved | | | chlorine molecules |

(c) Solid phosphorus(V) chloride contains $PCl_4^+$ ions.

(i) Draw a dot-and-cross diagram of a $PCl_4^+$ ion.

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Show only outer shell electrons.

**(1)**

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(ii) Predict the shape of a $PCl_4^+$ ion.
Justify your answer.

**(3)**

Shape .....
Justification

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**(Total for question = 16 marks)**

**Q5.**

**Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .**

Which of these has the **greatest** electrical conductivity?

☒ **A** $SF_6(g)$

☒ **B** $H_2O(l)$

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☒ **C** Hg(l)

☒ **D** $Na_2O(s)$

**(Total for question = 1 mark)**

**Q6.**

**Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Some physical properties of five substances are shown.
The letters are **not** element symbols.

| Substance | Electrical conductivity | | | Melting temperature /°C |
|---|---|---|---|---|
| | Solid | Liquid | Solution in water | |
| L | poor | good | good | 770 |
| M | good | good | reacts | 98 |
| N | good | good | insoluble | 1083 |
| P | poor | poor | insoluble | 113 |
| Q | poor | poor | good | 10 |

(a) Which of these substances could be metals?

☒ **A** N only
☒ **B** L and M only
☒ **C** M and N only
☒ **D** L, M and N only

(b) Which substance has properties showing that it changes from a molecular structure to ions when it dissolves in water?

☒ **A** L
☒ **B** M
☒ **C** P
☒ **D** Q

**(Total for question = 2 marks)**

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**Q7.**

This question is about the bonding in the elements of Period 3 in the Periodic Table.

The melting temperatures of the Period 3 elements are shown in the table.

| Element | Na | Mg | Al | Si | P | S | Cl | Ar |
|---|---|---|---|---|---|---|---|---|
| Melting temperature / °C | 98 | 650 | 660 | 1423 | 44 | 120 | −101 | −189 |

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(a) Sodium, magnesium and aluminium are metals.

(i) State what is meant by metallic bonding.

**(1)**

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(ii) Melting temperature depends on the strength of metallic bonding.
Explain why the metallic bonding in magnesium is much stronger than that in sodium.

**(3)**

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(b) (i) In the elements silicon, phosphorus, sulfur and chlorine, the atoms are joined by covalent bonds.
Describe the attraction between the atoms in a covalent bond.

**(1)**

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(ii) Explain why the melting temperature of silicon is much higher than that of phosphorus, by referring to their structures.

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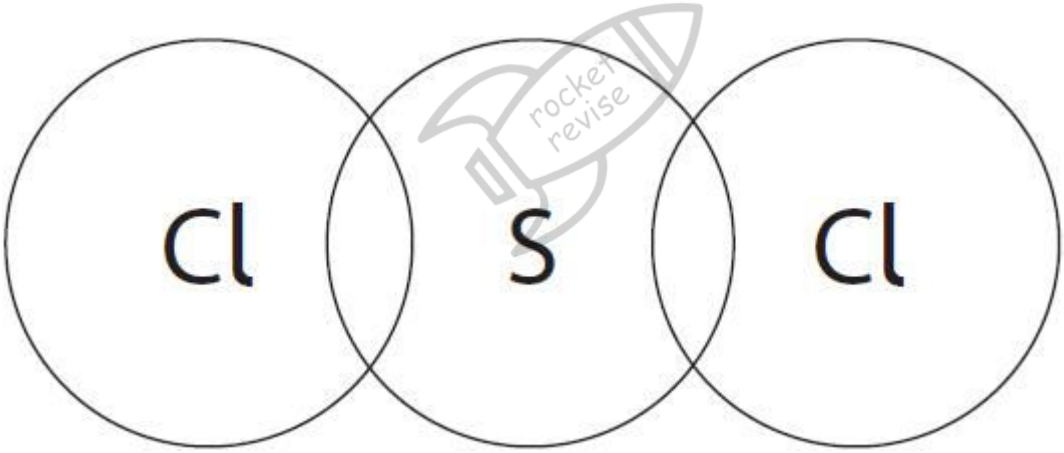
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(c) Sulfur reacts with chlorine to form sulfur dichloride, $SCl_2$.

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(i) Complete the dot-and-cross diagram of a molecule of sulfur dichloride. Use dots (•) for the chlorine electrons and crosses (×) for the sulfur electrons. Show the outer shell electrons only.

**(2)**




(ii) Suggest a value for the Cl—S—Cl bond angle. Justify your answer.

**(3)**

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**(Total for question = 13 marks)**

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**Q8.**

This question is about aluminium and its compounds.

(a) (i) State the shape of a molecule of aluminium trichloride, $AlCl_3$.

**(1)**

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(ii) Aluminium chloride can exist as a dimer formed from two molecules of aluminium trichloride.
Draw a diagram of this dimer, showing the bonding between the two molecules.

**(2)**

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(b) Aluminium trichloride reacts vigorously with water to produce hydrogen chloride and aluminium hydroxide, $Al(OH)_3$.

Write the equation for this reaction.
State symbols are not required.

**(1)**

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(c) Aluminium trichloride also reacts with sodium hydroxide.

$$AlCl_3(g) + 3NaOH(aq) \rightarrow Al(OH)_3(s) + 3NaCl(aq)$$

Calculate the maximum mass of aluminium hydroxide precipitated when excess aluminium trichloride reacts with 150 $cm^3$ of 1.5 mol $dm^{-3}$ sodium hydroxide.

**(3)**

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(d) (i) Describe the metallic bonding in aluminium.

**(3)**

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(ii) Explain how metallic bonding leads to the properties of electrical conductivity and malleability shown by aluminium.

**(2)**

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**(Total for question = 12 marks)**

**Q9.**

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**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

The melting temperature of beryllium is greater than that of barium.

What is the best explanation for this statement?

- ☒ **A** beryllium ions are smaller than barium ions
- ☒ **B** beryllium atoms have fewer outer shell electrons than barium atoms
- ☒ **C** beryllium ions have a smaller charge density than barium ions
- ☒ **D** beryllium atoms have a higher electronegativity than barium atoms

**(Total for question = 1 mark)**

**Q10.**

Magnesium is a metal in Group 2 of the Periodic Table. It reacts with chlorine to form the salt magnesium chloride, $MgCl_2$.

(a) Draw a dot-and-cross diagram for magnesium chloride.

Show outer shell electrons only.

**(1)**

(b) Magnesium conducts electricity when it is in the solid state. Magnesium chloride conducts electricity when it is molten or dissolved in water but not when it is in the solid state.

Explain these observations.

**(3)**

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(c) Magnesium chloride can also be made by reacting magnesium oxide with dilute hydrochloric acid.

$$MgO(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l)$$

(i) Write the **ionic** equation, including state symbols, for this reaction.

**(1)**

(ii) Calculate the minimum volume of 2.00 mol $dm^{-3}$ hydrochloric acid needed to completely react with 2.45 g of magnesium oxide.

**(3)**

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Minimum volume of hydrochloric acid = ..... $cm^3$

(d) A further method for making magnesium chloride is by reacting magnesium carbonate with dilute hydrochloric acid.

$$MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l) + CO_2(g)$$

Calculate the maximum mass of magnesium chloride that could be formed when 2.25 g of magnesium carbonate is added to excess dilute hydrochloric acid.

**(2)**

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Maximum mass magnesium chloride = ..... g

(e) Explain why the reaction to make magnesium chloride from magnesium oxide has a higher atom economy than the reaction using magnesium carbonate.

No calculation is required.

**(2)**

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**(Total for question = 12 marks)**

**Q11.**

**Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).**

Which row of the table shows properties consistent with the type of bonding shown?

| | | Bonding | Solubility in water | Melting temperature | Electrical conductivity of solid |
|---|---|---|---|---|---|
| ☐ | A | ionic | soluble | high | poor |
| ☐ | B | ionic | soluble | low | good |
| ☐ | C | metallic | soluble | low | good |
| ☐ | D | metallic | insoluble | high | poor |

**(Total for question = 1 mark)**

**Q12.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which are the correct bonding and structure for one of the substances listed?

| | | Substance | Bonding | Structure |
|---|---|---|---|---|
| ☐ | A | copper(II) sulfate | covalent | giant |
| ☐ | B | graphene | covalent | simple molecular |
| ☐ | C | iodine | metallic | simple molecular |
| ☐ | D | sodium | metallic | giant |

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**(Total for question = 1 mark)**

**Q13.**

This question is about silver.

(a) Silver chloride decomposes in light. This reaction was used in the first photographic plates.

Write an equation for this decomposition.
Include state symbols.

**(2)**

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(b) Silver can form in displacement reactions.

(i) State **two** observations that are made when copper metal is added to silver nitrate solution.

**(2)**

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(ii) Write an equation for this displacement reaction.
Include state symbols.

**(2)**

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(c) (i) Describe the bonding in pure silver metal.

Include a 2D diagram to show the arrangement of 12 of the silver particles involved.

**(3)**

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(ii) Pure silver cups are too soft so small amounts of copper are added to make an alloy.
Explain why copper makes the silver less malleable.

**(2)**

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**(Total for question = 11 marks)**

**Q14.**

This question is about magnesium, magnesium oxide and magnesium sulfate.

(a) A sample of magnesium contains three isotopes and has a relative atomic mass of 24.32.

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The table gives the relative abundances of two of these isotopes.

| Mass number | 24 | 25 |
|---|---|---|
| Relative abundance / % | 78.99 | 10.00 |

(i) Calculate the relative abundance and hence the mass number of the third isotope.
Give your answer to the appropriate number of significant figures.
You must show all your working.

**(4)**

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(ii) State **one** similarity and **one** difference between these isotopes.

**(1)**

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(iii) State which of these isotopes would be deflected most in a mass spectrometer. Justify your answer.

**(1)**

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(b) Magnesium oxide and magnesium sulfate are ionic compounds.

(i) Draw a dot-and-cross diagram to show the bonding in magnesium oxide, MgO.
Show outer electrons only.

**(2)**

(ii) The melting temperatures of magnesium oxide and magnesium sulfate are 2852°C and 1124°C respectively.
Explain why the melting temperature of magnesium oxide is significantly higher than that of magnesium sulfate.

**(2)**

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(c) The table gives some data about the electrical conductivity of magnesium and magnesium oxide.

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| | Electrical conductivity | |
|---|---|---|
| State | Magnesium | Magnesium oxide |
| solid | high | low |
| liquid | high | high |

Explain the similarities and differences in the electrical conductivity of the two substances.

**(2)**

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(d) Magnesium sulfate can be made by reacting magnesium with dilute sulfuric acid.

(i) Write an equation for the reaction that occurs.
Include state symbols in your answer.

**(2)**

(ii) Give **two** observations you would make when the reaction is taking place.

**(2)**

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(e) Hydrated crystals of magnesium sulfate, $MgSO_4 \cdot 7H_2O$ , can be made by reacting magnesium with sulfuric acid.

In an experiment, magnesium was added to 30.0 $cm^3$ of 0.500 mol $dm^{-3}$ sulfuric acid.
[$M_r$ value: $MgSO_4 \cdot 7H_2O$ = 246.4 $A_r$ value: Mg = 24.3]

(i) Calculate the number of moles of sulfuric acid used in this experiment.

**(1)**

(ii) Calculate the mass of magnesium needed to react with the sulfuric acid.

**(1)**

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(iii) Give a reason why slightly more than this mass of magnesium was used.

**(1)**

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(iv) State how the magnesium sulfate solution could be separated from the mixture produced in this experiment.

**(1)**

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(v) The magnesium sulfate solution was allowed to crystallise.
The crystals were dried and weighed.
The mass of the hydrated crystals, $MgSO_4 \cdot 7H_2O$, was 2.78 g.
Calculate the percentage yield in this experiment.

**(2)**

**(Total for question = 22 marks)**

**Q15.**

This question is about structure and bonding.

(a) Ionic bonding occurs between metals and non-metals.

(i) Describe ionic bonding.

**(1)**

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(ii) Explain why the ionic bonding in magnesium fluoride might be expected to be stronger than the ionic bonding in sodium fluoride.

**(2)**

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(iii) Sodium and oxygen react to form sodium oxide, which is also ionic. Draw the dot-and-cross diagram showing the ions present in sodium oxide.

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(2)

(iv) Draw the ionic lattice for sodium chloride.
You should show at least nine ions.

(2)

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(b) Non-metal atoms join together using covalent bonds.

(i) Describe covalent bonding.

(1)

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(ii) Methane, ammonia and water are covalently bonded compounds.
Draw the dot-and-cross diagrams for the molecules of methane ($CH_4$), ammonia ($NH_3$) and water ($H_2O$).

(3)

| methane | ammonia | water |
| --- | --- | --- |
| | | |

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(iii) The three molecules (methane, ammonia and water) each have a different shape. Complete the table below.

**(3)**

| molecule | shape | bond angle |
|---|---|---|
| methane | | |
| ammonia | | |
| water | | |

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(iv) Explain the differences between the molecular shapes and bond angles for ammonia and water.

**(2)**

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(v) Ethene and poly(ethene) have different melting temperatures.

| Molecule | Melting temperature (K) |
|---|---|
| ethene | 104.1 |
| poly(ethene) | 400 |

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Explain why their melting temperatures are different.

**(2)**

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(c) Metals are held together by metallic bonding.

(i) Describe metallic bonding.

**(1)**

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(ii) Metals conduct electricity when solid.
Explain, with the aid of a diagram, why copper conducts electricity.

**(2)**

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**(Total for question = 21 marks)**

**Q16.**

**Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.**

Which substance does **not** exist as a giant lattice?

- ☐ **A** silver (Ag)
- ☐ **B** sodium chloride (NaCl)
- ☐ **C** graphite (C)
- ☐ **D** buckminsterfullerene ($C_{60}$)

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**(Total for question = 1 mark)**

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